CANCER

By

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INTENDED LEARNING OUTCOME

- Identify risk factors of cancer
- Know methods of preventing cancer

- **Cancer is a most fearful disease, next to AIDS and Rabies. It is characterized by the following features:**
  - **Abnormal and uncontrolled growth of the cells.**
  - **The presence of aberrations in the nucleus.**
  - **Ability to invade the surrounding tissues and even distant organs (metastasis) later.**
  - **Eventual death of the person, if the tumor has progressed beyond a certain stage at which it can be successfully removed.**
  - **Cancer can occur at any site, at any time, in any individual.**

MAGNITUDE OF THE PROBLEM

- In 2022, there were about 20 million new cancer cases worldwide and 9.7 million deaths.
- Cancer is one of the leading causes of morbidity and mortality globally.
- By 2050, new cases are projected to rise to 35 million, driven largely by population growth and aging

Most common cancer in all patients

breast
lung
colon and rectum
prostate
skin (non-melanoma)
Stomach

Most common cancer in males

Lung
Prostate
Colorectal
stomach
liver cancer

Most common cancer in in females

breast
colorectal
lung
cervical
thyroid cancer

Most common causes of cancer death

lung
colon and rectum
liver
stomach
breast.

Age incidence:

- **Elderly predominance** → Cancer is more common in older adults due to cumulative genetic mutations and longer exposure to risk factors.
- **Developing** countries → Higher prevalence among younger people because populations are "demographically young."
- **Bimodal** age curve → Some cancers peak in both young adults and the elderly e.g. Leukemia, Hodgkin's disease, Kidney cancer
- **Familial** cancers → When cancer runs in families, onset tends to occur at an earlier age.
- **Cervical** cancer → Age of onset is linked to age at marriage; earlier marriage often correlates with earlier development of cervical cancer.

- **Sex:**
  - **Overall** → Cancer is generally **more** frequent in **men** than women.
  - Exceptions → **Women** have **higher** incidence in **reproductive organ cancers, thyroid, and gallbladder.**
  - **Reproductive** system cancers:
    - **Women** → **~50%** of all cancers are reproductive system-related (breast, cervical, ovarian, uterine).
    - **Men** → **Only ~15%** of cancers involve reproductive organs (prostate, testicular, penile).

CAUSES OF CANCER

- Cancer develops through a **multi-stage process**, starting from normal cells → pre-cancerous lesions → malignant tumors.
- It results from the **interaction** of genetic predisposition with external agents.
- Main categories of carcinogens:
  - **Physical** → ultraviolet and ionizing radiation.
  - **Chemical** → tobacco smoke, asbestos, aflatoxin, arsenic.
  - **Biological** → viruses (HPV, hepatitis), bacteria (H. pylori), parasites (Schistosoma).

- **Cancer develops gradually as a result of a complex mix of risk factors include lifestyle (smoking, alcohol, diet, obesity), environmental exposures, and heredity.**
- **About 80 percent of all cancers are related to the use of tobacco products, to what we eat and drink, or, to a lesser extent, to exposure to radiation or cancer-causing agents (carcinogens) in the environment and the workplace.**
- **Individual sensitivity varies → some people are more vulnerable to these factors than others.**

MAJOR RISK FACTORS FOR CANCERS

- **1. Tobacco**
  - Leading cause of cancer deaths worldwide.
  - Strongly linked to lung cancer (RR 10–20), plus cancers of larynx, oral cavity, pharynx, esophagus, pancreas, kidney, bladder.
  - Passive smoking increase lung/laryngeal cancer risk.
  - Smokeless tobacco lead to oral/head & neck cancers.

- **2. Alcohol**
  - Causes **oral**, **esophageal**, **liver**, **breast cancers; probable link to colon/rectum.**
  - Accounts for about **3%** of cancers (4% men, 2% women).
- **3. Occupational Exposures**
  - About **25 chemicals confirmed carcinogens** (e.g., asbestos, nickel, cadmium, radon, benzene).
- **4. Environmental Pollution**
  - Air, water, soil contaminants (asbestos, chlorination by-products).
  - Contributes to about **1–4%** of cancers in developed countries.

- **5. Food Contaminants**
  - **Aflatoxins** (fungal toxins in maize/groundnuts) lead to hepatocellular carcinoma.
  - **Pesticides/organics** linked to pancreatic, breast, lymphoma, leukemia.
- **6. Radiation**
  - **Ionizing** radiation leads to leukemia, breast, thyroid cancers.
  - **Electromagnetic** fields leads to childhood leukemia (possible).

- **7. Sunlight (UV radiation)**
  - **Causes skin cancers.**
- **8. Chronic Infections**
  - **Account for about 18% of cancers worldwide.**
  - **Examples: HBV/HCV (liver), Human Papilloma Viruses (HPV) (cervix), Epstein-Barr virus (EBV) (lymphoid), H. pylori (stomach), Schistosoma (urinary bladder).**

- **9. Diet & Nutrition**
  - About 30% of cancers linked to diet.
  - Western diet increase risk of colorectal, breast, prostate, endometrial cancers.
  - Fruits/vegetables protective (decrease risk (protect against digestive tract cancers).
  - High salt/preserved foods increase risk of gastric cancer.
  - Red/processed meat increase risk of colorectal cancer.
  - High fat intake increase risk of colorectal cancer.
  - Protective micronutrients: vitamins A, C, E, folate.

- **10. Obesity & Physical Inactivity**
  - Western diet and sedentary lifestyle increase risk of colon, breast, prostate, endometrial cancers.
  - Strong risk factor for endometrial and postmenopausal breast cancer.
- **11. Genetic Susceptibility**
  - Familial clustering (breast, ovary, colon, melanoma).
  - Inherited syndromes (e.g., BRCA1, retinoblastoma).
  - Accounts for about 4% of cancers.
- **12. Reproductive Factors & Hormones**
  - Female hormones influence breast, ovarian, endometrial cancers.
  - Oral contraceptives: slight increased risk of breast cancer risk but decrease risk of ovarian/endometrial risk.
  - Hormone replacement therapy: increase risk of breast/endometrial cancer risk.

CANCER PREVENTION:

- **Cancer prevention: By elimination of or protection against factors known or believed to be involved in carcinogenesis**

Primary prevention:

- **Health promotion:**
  - **Encourage research for etiology, risk factors diagnosis.**
  - **Health education for public.**
  - **Monitoring for cancer registry.**

Specific protection:

- not using tobacco;
- maintaining a healthy body weight;
- eating a healthy diet, including fruit and vegetables and low in fat;
- doing physical activity on a regular basis;
- avoiding harmful use of alcohol;
- getting vaccinated against HPV and hepatitis B if vaccination is recommended;
- avoiding ultraviolet radiation (sun exposure).
- reducing exposure to ionizing radiation (occupational or medical diagnostic imaging);
- reducing exposure to outdoor and indoor air pollution.
- using low estrogen contraceptives when needed.
- treatment of pre-cancerous lesions.

SECONDARY PREVENTION:

- **Early Diagnosis and Treatment:** Cancer mortality can be reduced if cases are detected and treated early. Early detection can done by:
- **Identification of high risk groups**
  - Occupation.
  - Family history.
  - Environmental factors.
  - Certain risk factors.
  - Pre-cancerous condition.

Screening and case finding: by

1. **A self examination:** (after age 30) → Individuals should check themselves and seek medical advice if warning signs appear, such as
   - A **lump** or hard area in the breast or testes
   - A **change** in a wart or mole
   - A **persistent change** in digestive or bowel habits
   - **Persistent** indigestion or difficulty in swallowing
   - **Persistent** cough or hoarseness
   - **Excessive or irregular** menstrual blood loss
   - **Blood loss** from any natural orifice
   - A **swelling** or sore that does not get better
   - **Unexplained** loss of weight.

2. Clinical examination → Doctors can detect ~75% of cancers through thorough physical exams, especially in accessible body sites. A lump or hard area in the breast or testes

- **Screening techniques:**
  - **Exfoliative cytology** → Pap test, cytopipette irrigation for cervical cancer (highly effective in early detection).
  - **Imaging** → X-rays, ultrasound for internal cancers.
  - **Thermography** → Detects abnormal heat patterns in skin over cancer sites.
  - **Endoscopy** → Bronchoscopy, colonoscopy, gastroscopy, proctosigmoidoscopy for internal organ cancers.
- **Mass screening** of all men above 50 and women above 30, at least in selected groups, should be undertaken.

TREATMENT:

- Cancer is **treated** with surgery, radiation therapy, chemotherapy, hormone therapy, or biological therapy.
- Patients with cancer are often treated by a **team** of specialists, which may include a medical oncologist (specialist in cancer treatment), a surgeon, a radiation oncologist (specialist in radiation therapy), and others.

TERTIARY PREVENTION

- **Disability limitation:**
  - This is done by giving intensive treatment to prevent the development of further disability.
- **Rehabilitation:**
  - This is given for those who have undergone surgery such as amputation of leg, colostomy, mastectomy, etc. and are rehabilitated with a prosthesis and training, later placed in a suitable job.

Thank you